

Chapter I: Architecture of Programmable Logic Controller (PLC)

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Chapter I: Architecture of Programmable Logic Controller

1 Introduction

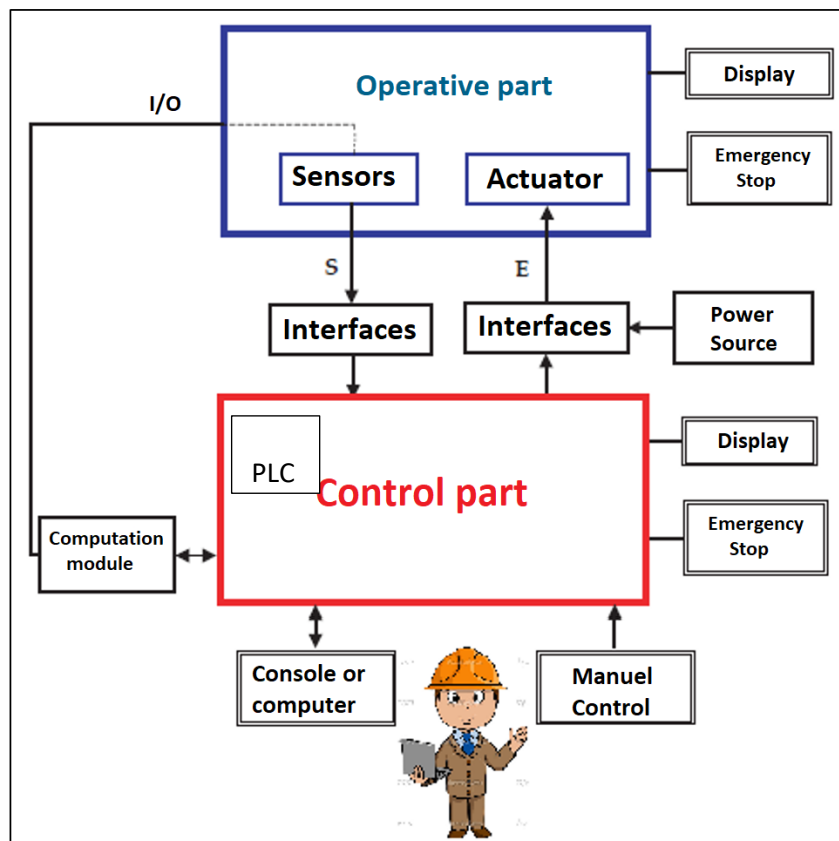
Automation involves integrating production means with automatic control systems to reduce or even eliminate human intervention. It enables companies to enhance their competitiveness by optimizing product costs, improving quality, and more.

2 Definition of an Automated System

A production system is considered automated when it can autonomously manage a predefined work cycle, which is divided into sequences and/or steps. Automated systems, widely used in the industrial sector, share a common basic structure (Figure 1.1). They are generally composed of two interconnected and complex parts:

- The operative part (PO)
- The control part (PC)

These two parts communicate with each other through an interface. There is also a relational component that facilitates dialogue between the system user and the automated system through the control part.



*optional

Figure 1.1. Automated Process

3 Description of the Different Parts

3.1 The Operative Part

This is the visible part of the system. It includes the process elements, which are: **sensors, pre-actuators, and actuators.**

3.1.1 Sensors:

A sensor is characterized by several criteria, such as:

- The observed physical quantity,
- Sensitivity,
- Precision,
- Measurement range,
- Linearity,
- Bandwidth,
- Operating temperature range,
- Resolution, etc.

Sensors can be classified according to several criteria:

- **Energy supply:** Passive sensors or active sensors.
- **Output type:** Analog sensors or digital sensors.

The figure below illustrates some types of sensors, such as: **liquid level sensor, shock detector, humidity sensor, limit switch, ultrasonic proximity sensor, gas detector, photoelectric cell, push button, switch, and emergency stop button.**



Figure 1.2. Different Types of Sensors

3.1.2 Actuators:

Actuators convert the received energy into a physical phenomenon that performs work (heat generation, light emission, movement, etc.). Automated systems use various types of actuators, which can be classified according to two criteria:

- **The type of energy supply:** Electrical, pneumatic, or hydraulic.
- **The physical phenomenon produced:** Heat, movement, light, sound, etc.

The figure below illustrates some types of actuators, such as: **cylinder, rotary actuator, stepper motor, direct current motor, buzzer, indicator light, solenoid valve, heating resistors, 7-segment display, and fan.**

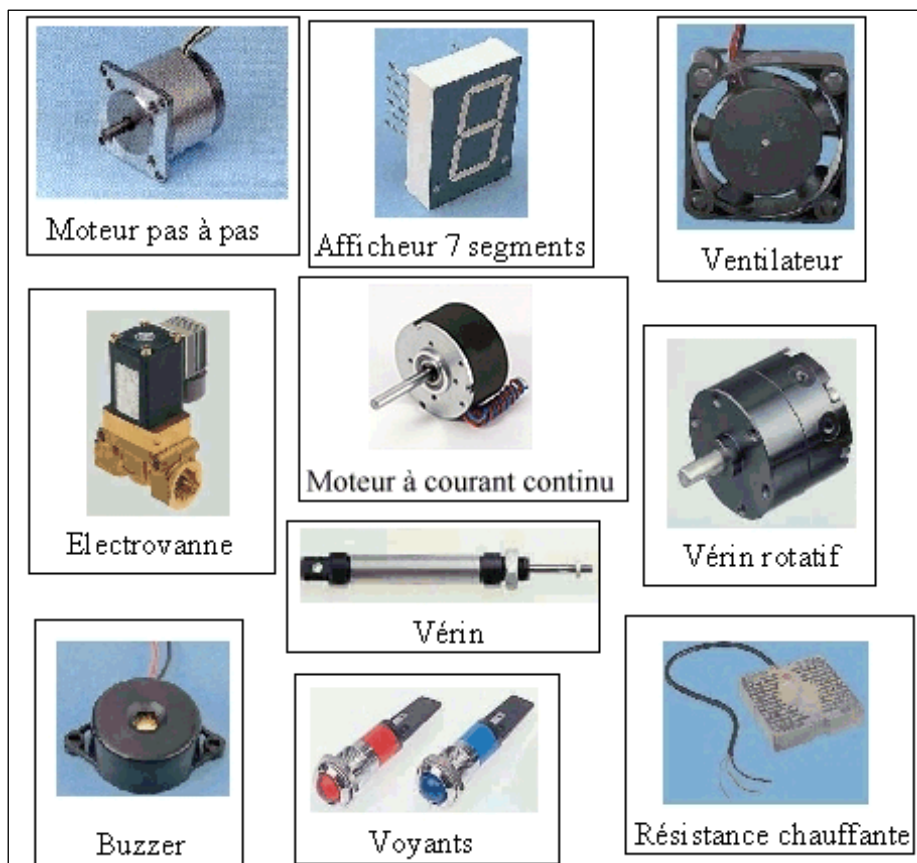


Figure 1.3. Different Types of Actuators

3.1.3 Pre-Actuators:

A pre-actuator is a component that receives commands from the control part (PC). Its role is to distribute the necessary and appropriate energy to the actuator, as the PC is generally incapable of directly providing the energy required by the actuator.

The choice of a pre-actuator depends on the characteristics of the control part, the energy source, and the actuator. If the actuator is electrical, the pre-actuator will also be electrical (e.g., contactor, relay, power card, variable speed drive, chopper), as shown in the figure below. In cases where the actuator is pneumatic, the pre-actuator will be pneumatic (e.g., distributor).



Figure 1.4. Different Types of Pre-Actuators.

3.2 The Control Part

The field of automation logically manages the orderly sequence of operations to be performed. It receives information from the sensors in the operative part and sends it back to the same operative part, directed towards the pre-actuators and actuators.

3.3 The Interface

The control part and the operative part differ in nature. To ensure proper communication between these two parts, an object called the "Interface" is used.

These interfaces act as "translators" that connect the control part to the operative part.

3.4 The Relational Part (RP)

This part typically includes a Human-Machine Interface (HMI) for interacting with the control system, such as a control panel. The panel is equipped with control elements allowing: emergency stop, manual control of actuators, powering on/off the installation, selecting operating modes (start/stop), and setting references. It also includes various signaling components, such as screens, indicator lights, displays, buzzers, alarms, etc.

4 Information and Energy Chains

4.1 The Information Chain

The information chain performs the following functions:

- **Acquiring data** about the state of a product, from the HMI, or a process managed by other systems.
- **Processing the data.**
- **Communicating the data processed by the control unit (usually a Programmable Logic Controller, PLC)** to issue commands for the energy chain and/or to send messages to the HMI or other information chains.

4.2 The Energy Chain

The energy chain comprises the following generic functions: **Supply**, **Distribution** (handled by the pre-actuator), **Conversion** (handled by the actuator), and **Transmission**, all of which contribute to executing an action. The primary role of the energy chain is to ensure the execution of a service function, with its characteristics specified in the design specifications.

The figure below illustrates the energy and information chains of an Automated Production System (APS).

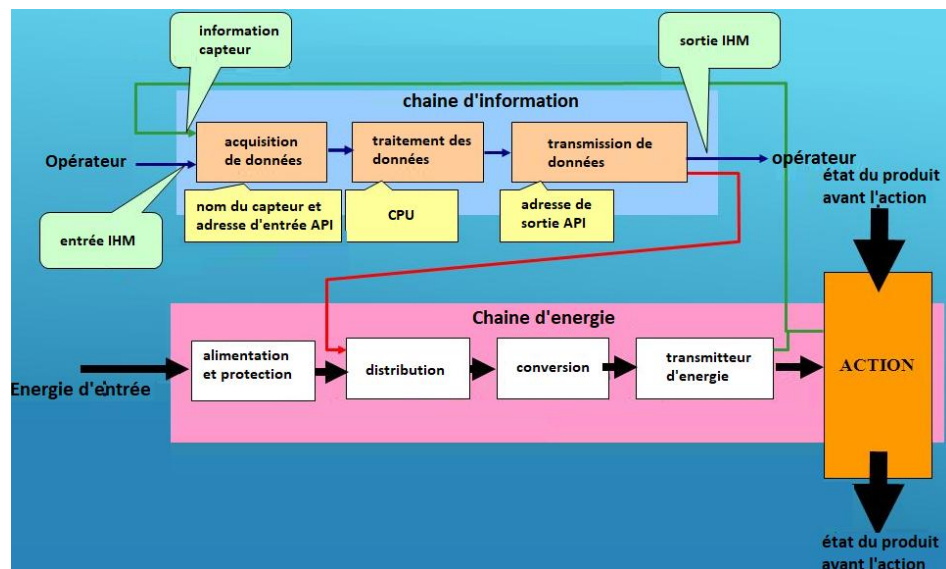


Figure 1.5. Overview of Generic Functions in the Energy and Information Chains of an Automated Production System

4.3 Example of Energy Chain Equivalence

The figure below illustrates the equivalence between two different energy chains: one uses electrical energy, while the other uses pneumatic energy.

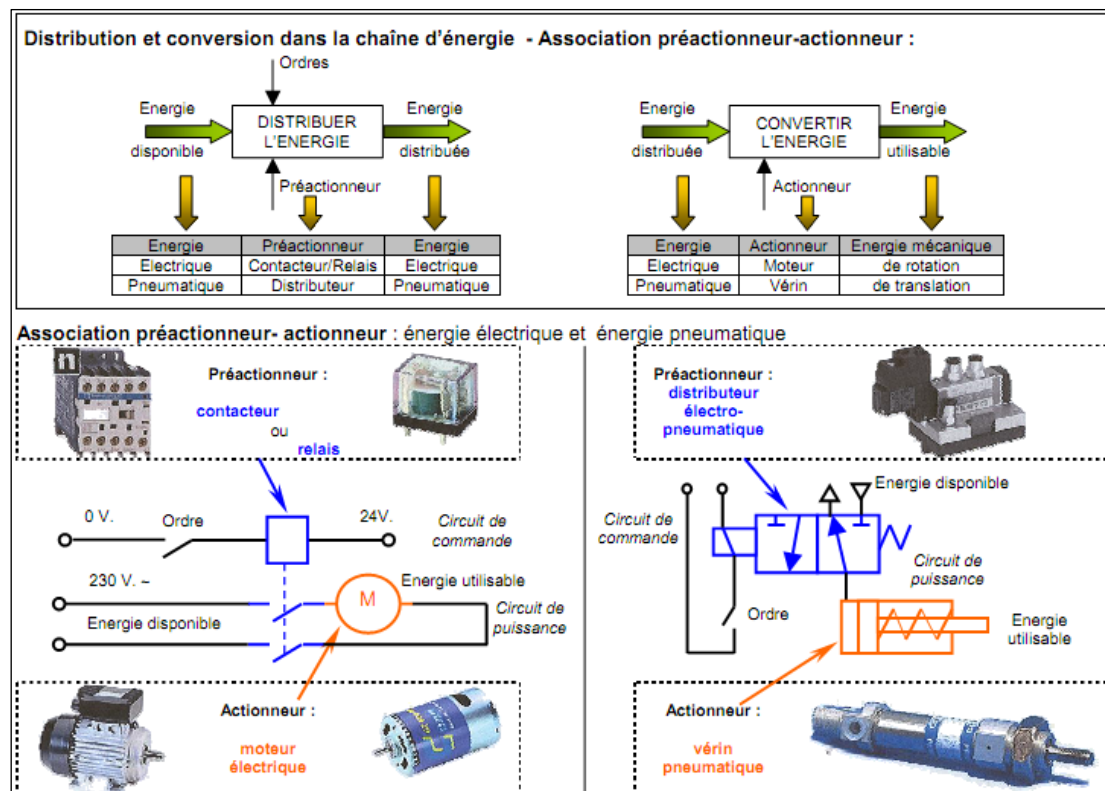


Figure 1.6. Diagram of Equivalence Between an Electrical Energy Chain and a Pneumatic Energy Chain

5 Different Types of Control Systems

5.1 The Combinational Automated System

These systems do not use any memory mechanism: for a given combination of inputs, there is only one corresponding combination of outputs. The associated logic is **combinational logic**, and the tools used for their design include Boolean algebra, truth tables, and Karnaugh maps. Automated systems using "combinational" techniques are rarely employed today. However, they can still be used for simple mechanisms with a limited number of actions to perform. They also have the advantage of requiring very few components.

5.2 The Sequential Automated System

These systems are the most widespread in the industrial domain. The cycle progresses step by step. For a given input situation, several output situations may correspond. The selection of one step or another depends on the previous state of the device.

5.3 Programmed Logic: Electrical Control

The main component is called the Programmable Logic Controller (PLC). Detection is electrical. The control of actuators is carried out through relays or distributors. There are many brands of programmable logic controllers available on the market, including Telemecanique, Siemens, Omron, Allen-Bradley, Cegetel, etc.

5.4 Wired Logic: Pneumatic Control

The primary component is called the **sequencer module**, and the combination of modules forms a system called a sequencer. Detection is pneumatic, and the actuation of distributors is performed by compressed air acting on a piston, which moves the distributor spool to the right or left. This system, known as **all-pneumatic**, is both homogeneous and reliable.

5.5 Servomechanisms

In these systems, the objective is for the output to accurately follow the variations of the input, with minimal response time. Examples include power steering in automobiles or the control of an airplane's flight surfaces. Applications : industrial robots.

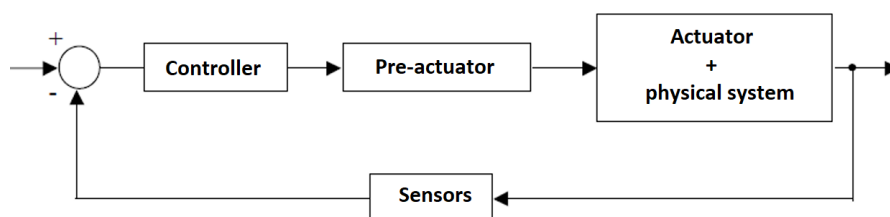


Figure 1.7. Functional Diagram of the Principle of a Servomechanism

6 Application Domains of Automated Systems

Today, it would be challenging to design a production system without relying on automated systems. These systems are found extensively in the field of industrial production and are used daily for various tasks. To operate and maintain these devices, technicians specializing in automation are employed, handling the installation and maintenance of each system.

Advantages of Automated Systems

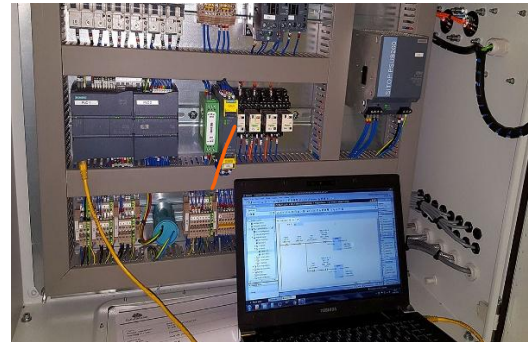
- Accelerated production capacity.
- Suitability for all production environments.
- Flexibility of use.
- Creation of jobs for automation specialists.

Disadvantages of Automated Systems

- High equipment costs, especially for hydraulic systems.
- Maintenance requires a structured approach.
- Job reduction in certain sectors.

7 Definition of a Programmable Logic Controller (PLC)

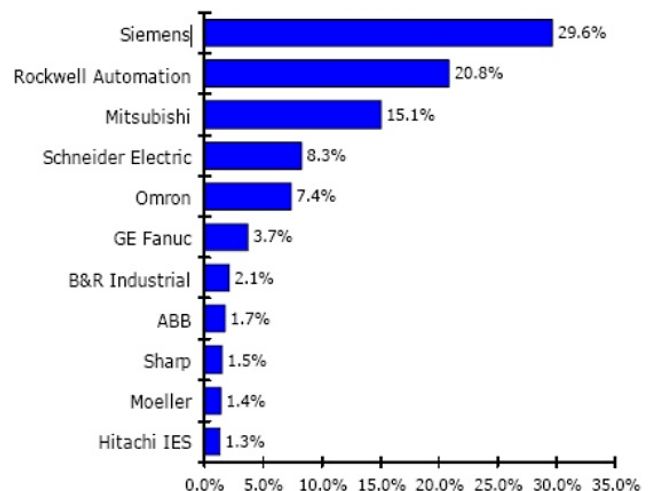
A Programmable Logic Controller (PLC) is an electronic programmable device designed for controlling industrial processes through sequential processing. It sends commands to pre-actuators (operative part or **PO**, on the actuator side) based on input data (sensors) (**Operative part or PO**, on the sensor side), instructions, and a computer program.



PLCs emerged in the United States around 1969 in response to the automotive industry's desire to develop automated manufacturing lines that could adapt to advancements in technology and changes in manufactured models.

8 Most Popular and Widely Used PLC in the Industrial Sector

<u>Américains :</u>	1. Allen Bradley 2. Gould Modicon 3. Texas Instruments 4. General Electric 5. Westinghouse 6. Cutter Hammer 7. Square D
<u>Européens :</u>	1. Siemens 2. Klockner & Mouller 3. Festo 4. Telemecanique
<u>japonais :</u>	1. Toshiba 2. Omron 3. Fanuc 4. Mitsubishi



9 Advantages and Disadvantages

9.1 Advantages

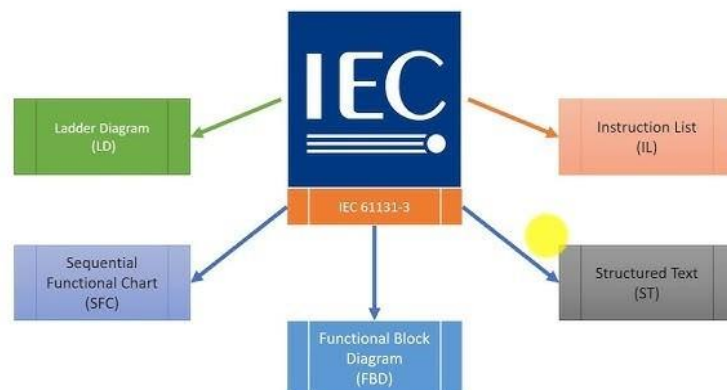
PLCs offer many benefits:

- The components are particularly robust, allowing them to operate in harsh environments (ambient dust, electromagnetic interference, vibrations, temperature variations, etc.).
- They feature optimized electronic circuits for interfacing with the physical inputs and outputs of the system, ensuring minimal execution time and effective real-time operation.

- Maintenance is easy, as modules can be replaced very quickly.

9.2 Disadvantages

- They are more expensive than traditional computing solutions, such as those based on microcontrollers.
- The cost depends on factors such as: the number of inputs/outputs required, the memory available for programming, and the inclusion of industry-specific modules.
- Training is required, as PLCs require mastery of specific programming languages. However, these languages are often perceived by users as more accessible and visual than traditional programming languages.
- The variety of brands and models results in a diversity of languages and variable mappings, despite the existence of the IEC 1131 standard.



10 Types of Programmable Logic Controllers (PLCs)

There are two types of PLCs:

- The **monoblock type**.
- The **modular type**.

10.1 Monoblock PLC

The monoblock type generally has a limited number of inputs and outputs, and its instruction set cannot be expanded. Although it is sometimes possible to add input/output extensions, the monoblock type is primarily designed for solving simple automation tasks based on sequential logic.

Example:



Figure 3.2. Monoblock PLC Modicon M221 by Schneider Electric

The Modicon M221 is a monoblock PLC by Schneider Electric.

Main Specifications:

- **Power supply voltage** : 24 V DC
- **Number of binary inputs**: 24, including 4 fast inputs compliant with IEC 61131-2 Type 1
- **Number of analog inputs**: 2 (0...10 V)
- **Output logic type** : Transistor
- **Number of binary outputs**: 16 transistors, including 2
- **Logical output voltage** : 24 V DC
- **Logical output current** : 0.5 A

10.2 Modular PLC

The modular type, on the other hand, is adaptable to all situations. Depending on the requirements, analog input/output modules are available, in addition to specialized modules such as **PID**, **BASIC**, and **C Language**, among others. The modularity of PLCs allows for quick troubleshooting and greater flexibility.

11 Structure of a PLC

A PLC is structured around:

- A **processing unit** or CPU (Control Processing Unit)
- A **power supply** (PS: Power Supply)
- **Modules** that vary depending on the needs of the application.

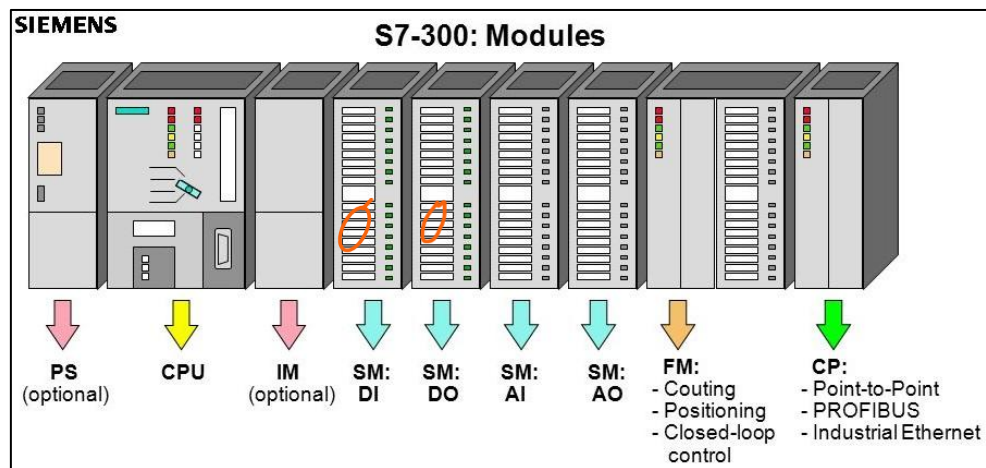


Figure 3.3. Structure of the Simatic S7-300 PLC by Siemens

11.1 Signal Modules (SM: Signal Modules)

Signal modules serve as the interface between the PLC and the process.

11.1.1 Binary Input/output Modules:

These modules allow the connection of binary signals (sensors and digital actuators) to the PLC.

a) Digital Input Modules (DI: Digital Input):

They are designed to receive information from sensors and adapt the signal by shaping it, eliminating noise, and electrically isolating the control unit from the operative part.

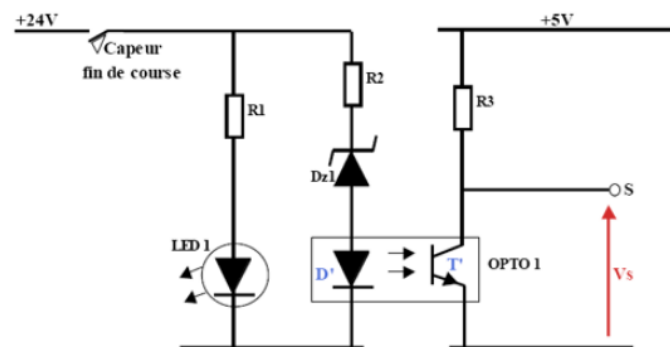


Figure 3.4. Example of a Typical Input Card for a PLC

b) Digital Output Modules (DO: Digital Output):

These modules are designed to control pre-actuators and signaling elements of the system. They adapt the voltage levels of the control unit to those of the operative part of the system while ensuring galvanic isolation between the two.

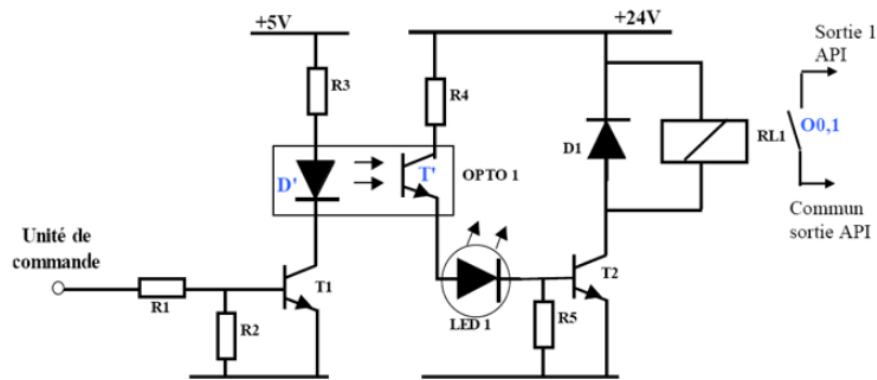


Figure 3.5. Example of a Typical Output Card for a PLC

11.2 Analog Input/Output Modules

These modules are connected to continuously varying signals.

- **Analog Input Modules (AI: Analog Input):** Designed to receive continuous signals.
- **Analog Output Modules (AO: Analog Output):** Designed to send continuous signals.

11.3 Communication Modules (CP: Communication Processor)

These modules operate under various protocols and were introduced to facilitate communication between:

- Multiple PLCs.
- A PLC and remote inputs/outputs.
- A PLC and Human-Machine Interfaces (HMI).

Examples of communication protocols: Modbus, Modbus Plus, Profibus, InterBus, RS232, RS-485, Ethernet, DeviceNet, LonWorks, FIPIO, FIPWAY, CANopen, etc.

Example: The ASi bus (Actuators Sensors interface) is a Master/Slave type sensor/actuator bus that connects up to 31 slaves (sensors/actuators) on a two-wire cable.

11.4 Interface Modules (IM: Interface Module)

Interface modules play a crucial role in automated systems, particularly in configurations where the PLC must manage inputs and outputs (I/O) across extended sites. These modules enable flexible expansion of available I/Os, integrating specialized functions such as variable speed drive control, high-speed counter management, and other domain-specific equipment (industry-specific modules).

Using interface modules as slaves in a decentralized peripheral system provides several advantages:

- Reduced wiring costs.
- Increased flexibility.

- Improved performance.

Example:

A beverage production line consisting of two units, filling and mixing, controlled by a single PLC (see figure below). Since the PLC is located far from the mixing unit, the use of longer cables becomes necessary, resulting in higher cabling costs.

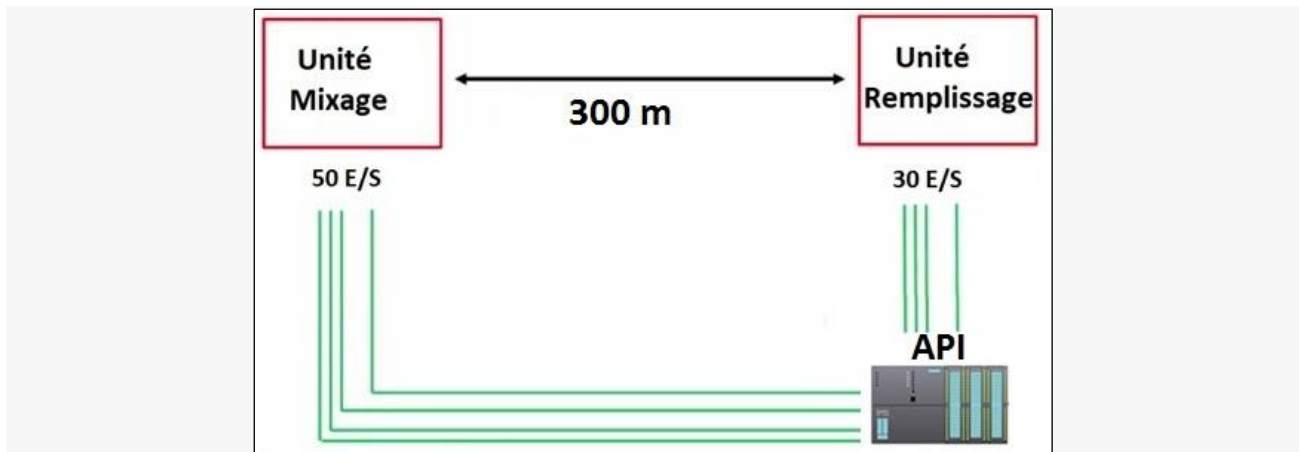


Figure 3.6. Wiring Diagram Between the PLC and Two Production Units

The solution to this problem is to use a **decentralized Input/Output (I/O) peripheral** placed near the mixing unit. By doing so, less cabling will be required, and maintenance operations will be easier to perform.

In this setup, communication between the control PLC and the decentralized I/O module is established via the **Profibus-DP protocol** (Decentralized Peripherals). The PLC acts as the master, while the decentralized I/O module functions as the slave.

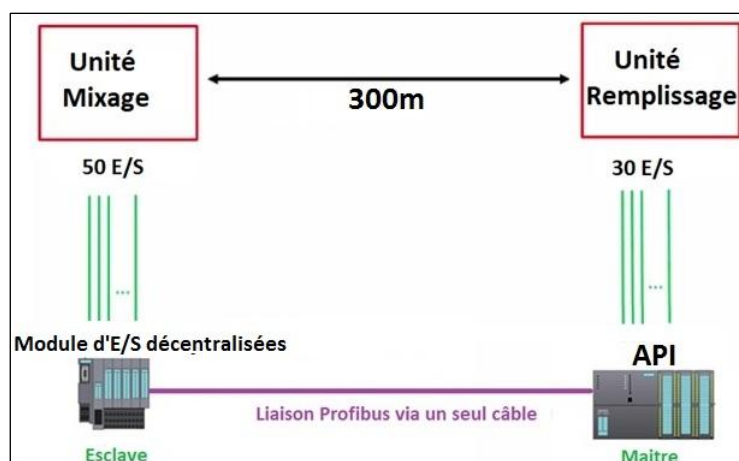


Figure 3.7. Wiring Diagram Between a PLC and a Decentralized I/O Module

11.5 Function Modules (FM: Function Module)

Also known as industry-specific modules, these are intelligent modules that independently execute technological tasks such as counting, measuring, cam control, PID control, and motion control. They

reduce the load on the processor and are used in applications requiring high precision and dynamic response, such as high-speed counting and weighing.

11.6 Local Human-Machine Interface Modules (HMI: Human-Machine Interface)

These include devices such as a control panel (touchscreen or with a keyboard) or a maintenance terminal. These are connected to the PLC via an industrial network (proprietary or open) and display messages or a representation of the process.

12 Internal Architecture of a PLC

The internal structure of a PLC is similar to that of a simple computer system. The central unit combines the processor and central memory, managing the interpretation and execution of program instructions. Instructions are executed sequentially, driven by a clock.

The PLC also includes input/output interfaces, and all its components are typically powered by a DC power supply unit.

Below, we define each of the elements that make up the PLC:

- **Processing Unit (CPU).**
- **Input/Output Interfaces.**
- **Power Supply:** 230 V, 50/60 Hz (AC) - 24 V (DC).

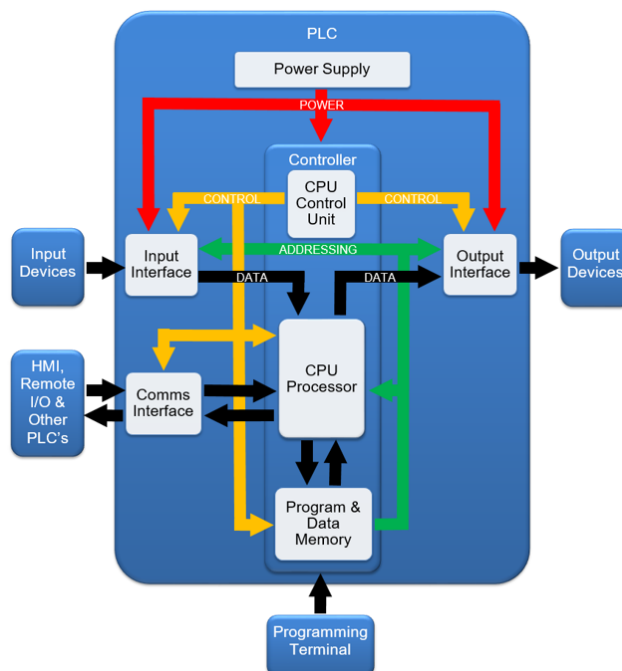


Figure 3.8. Internal Architecture of a PLC

12.1 The CPU

The CPU is often referred to as the brain of a Programmable Logic Controller (PLC) because it coordinates all the essential functions of the system. It executes the user program, manages inputs and outputs, processes data, and ensures communication with other devices or networks. The CPU typically consists of two main components: the processor and the memory, each playing a crucial role in the PLC's operations.

12.2 The Processor

The processor is the core of the CPU. It can be a microprocessor or a microcontroller. It oversees all system operations and performs all the necessary tasks (logical and arithmetic functions) required for program execution.

12.3 The Memory

The CPU memory in a PLC is essential for storing and retrieving the data and instructions necessary for system operation. This memory can be organized into different areas or segments, each serving a specific role in data processing and management. While the precise classification may vary depending on the manufacturer or the specific PLC model, the memory is generally divided into four main parts:

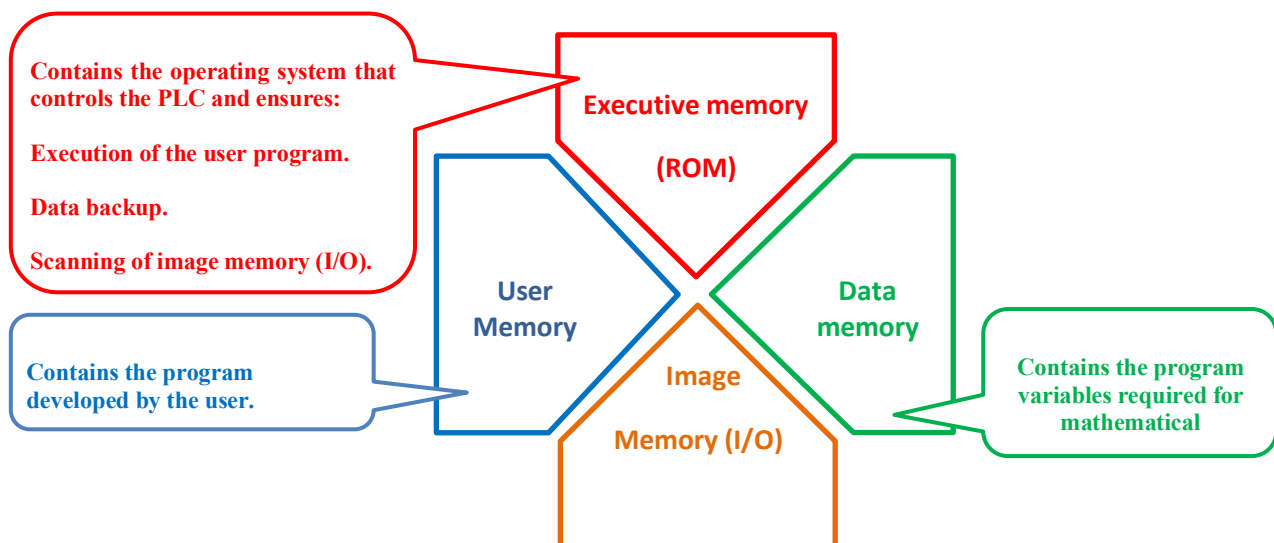


Figure 3.9 Organization of the CPU Memory

a. Input/Output Image Memory

- The **Input Image Memory (IIM)** consists of memory locations used to maintain the ON or OFF states of each input variable.

- The **Output Image Memory (OIM)** contains memory locations that store the ON or OFF states of the output variables. Data is saved in the OIM after executing the user program and waits to be transferred to the output modules.

b. Data Memory

Used to store numerical data required for mathematical calculations, logical functions, etc.

c. User Memory

Contains the user program.

d. Executive Memory

Used to store an executive program or system software. A PLC operating system is a special program that controls the CPU's actions and the execution of the user program. The PLC operating system scans the image memory, interprets, and executes the instructions of the user program stored in the main memory. It is provided by the PLC manufacturer and permanently stored in ROM.

12.4 Input/Output Interfaces

The interfaces of the PLC handle the reception and formatting of signals from external sources (sensors, push buttons, etc.) and the transmission of signals to external devices (control of pre-actuators, signaling lights, etc.). The design of these interfaces with galvanic isolation or optoelectronic decoupling ensures the PLC's protection.

12.5 Power Supply

The power supply provides power to the memory system, processor, and I/O modules. It converts high-level AC line voltage into various operational DC voltage levels.

13 Operating Principle of a PLC

PLC programs are processed in a specific cycle, typically as follows:

1. **Input scan:** Digital and analog inputs are read and stored in the Input Image Memory (IIM) to ensure a stable image of input states during the cycle.
2. **Program Processing:** The PLC executes the user program, processing input data to determine output actions.
3. **Output Update:** Program results are used to update the outputs by transferring data from the Output Image Memory (OIM) to the output modules.
4. **External Communications:** The PLC manages data exchanges with external devices to coordinate between PLCs, supervision systems, and other necessary communications.
5. **Self-Diagnosis:** Regular checks are performed to ensure proper operation of the PLC, with the capability to log diagnostics and report anomalies.

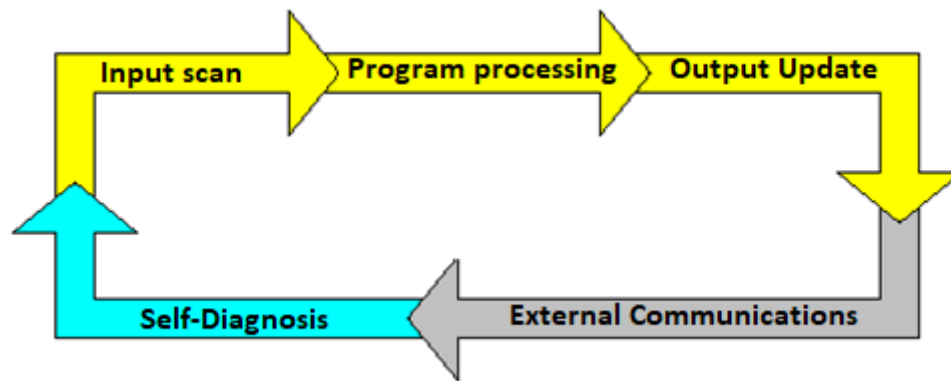


Figure 3.10. Operating Principle of a PLC

14 PLC Cycle Time

The cycle time of a PLC varies depending on:

1. **Program Size:** A longer and more complex program increases execution time due to the higher number of instructions and data to process.
2. **Calculation Complexity:** Operations requiring complex calculations extend the cycle time due to higher computational demands.
3. **Number of Inputs/Outputs:** A large number of I/O operations can slow the cycle, as each operation requires additional checks and communications with external devices.
4. **CPU Power:** The speed of the processor directly impacts the cycle time. A more powerful CPU reduces the time required to complete a cycle by executing instructions faster.
5. **Process Requirements:** The specific industrial process being controlled can dictate cycle time requirements, with some processes requiring near-instantaneous responses for safety and efficiency.

The cycle time typically ranges from a few milliseconds to tens of milliseconds.

Sensor readings and actuator commands are performed through polling.

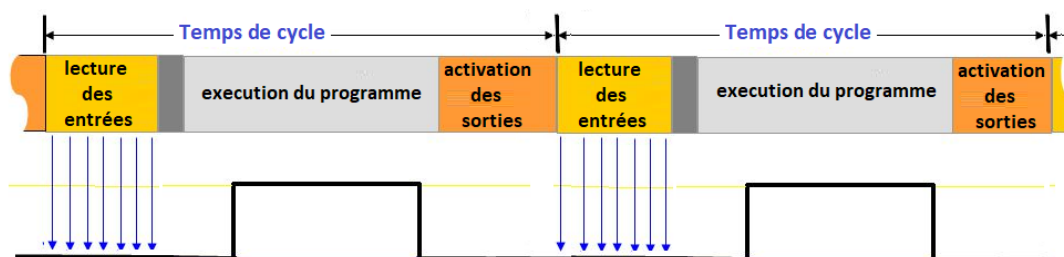


Figure 3.11. Execution Time of a Program in a PLC

15 Operating Principle of a Siemens Simatic S7 Program

Siemens PLCs are recognized as the most widely used control systems globally in the industrial sector. Their predominance is attributed to the reliability, flexibility, and advanced performance of Siemens PLCs, which effectively meet the complex demands of modern industrial environments.

In this context, we chose a Siemens PLC for our practical automation work. Specifically, we selected the **S7-1214C DC/DC/RLY** model from the **S7 1200 series**. The S7-1214C is designed to provide high flexibility and efficiency in managing industrial processes, with integrated inputs and outputs that support direct current (DC) for both inputs and outputs, as well as relays (RLY) for controlling external devices.

15.1 Main Organization Block (OB1):

Cyclic OBs (Organization Blocks) are fundamental to the structuring of Siemens PLCs. These code blocks play a crucial role in program organization, as they are processed cyclically, ensuring the sequential execution of automated tasks.

The repeated execution of the OB allows continuous monitoring and response to changes in the state of inputs and outputs, as well as variations in the internal data of the program. Within these blocks, a variety of logical, mathematical, or data management instructions can be programmed, defining the behavior of the automated system under various operational conditions.

15.2 Startup Organization Block:

The processing of startup OBs occurs once when the CPU transitions from **STOP** to **RUN** mode. The processing of the startup OB100 is followed by the cyclic OB (OB1).

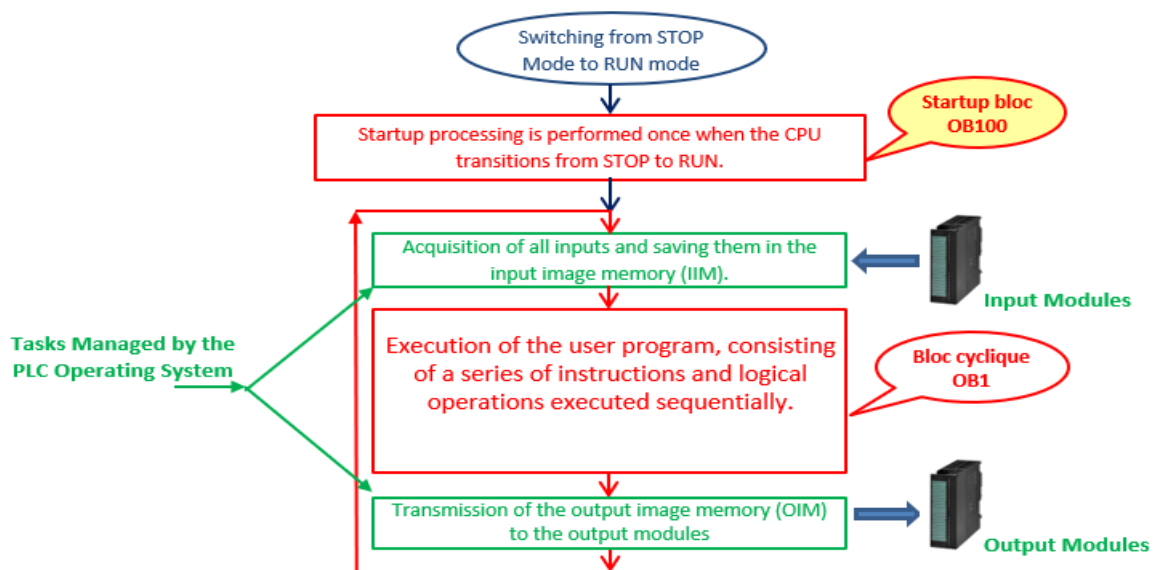


Figure 3.12. Operating Principle of a Simatic S7 Program

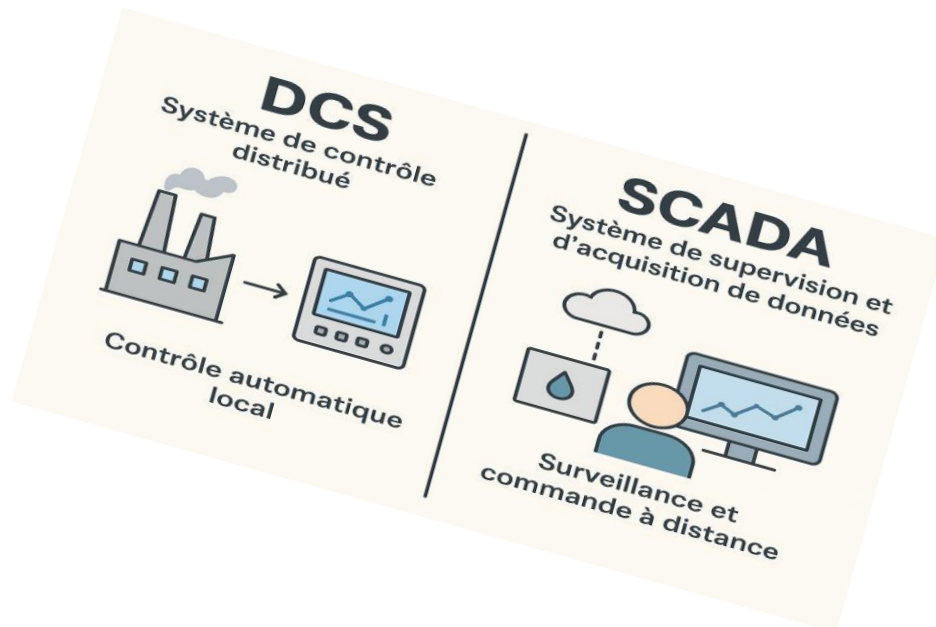
• Evolution and Structure of PLCs (Programmable Logic Controllers)

→ PLCs have evolved from the first simple models (to replace relays) into modern, more powerful systems capable of handling complex processes, network communication, and software integration.

→ Their basic structure includes: a processor, a power supply, input/output (I/O) modules, and communication modules.

Difference between PLC, DCS, and SCADA

- **PLC (Programmable Logic Controller):** Local system that directly controls a machine or a process.
- **DCS (Distributed Control System):** Distributed system used to control large-scale complex processes (chemical plants, power stations, etc.).
- **SCADA (Supervisory Control and Data Acquisition):** Supervision system that collects data from PLCs/DCSs and enables visualization, monitoring, and remote control.



System	Main role	Typical use	Example
PLC (Programmable Logic Controller)	Local control of a machine process	Production line, robot, convoyeur	Siemens S7-A200, Schneider M340
DCS (Distributed Control System)	Distributed control of complex processes	Refinery, power plant, chemical industry	ABB 800xA, Siemens PCS7
SCADA (Supervisory Control and Data Acquisition)	Supervision, data collection and display, remote control	Power grid, water management, large plants	WinCC (Siemens), Wonderware

- **Examples of PLCs by manufacturer**

Manufacturer	Old ranges	Recent ranges	Associated software
Siemens	S7-300, S7-400, WinAC	S7-1200, S7-1500	Step5, step7, Microwin, TIA Portal
Schneider	TSX57	M340, M580	PL7, Unity Pro

- **PLC structure (modules) :**

- **Rack:** Frame where the modules are installed.
- **Power Supply (PS):** Provides energy to the modules.
- **Digital I/O modules:** For reading or controlling binary signals (0/1).
- **Analog I/O modules:** For reading or controlling continuous values (temperature, pressure, speed, etc.).
- **Communication modules:** To connect the PLC to a network or to other devices.
- **Function modules:** Special-purpose cards (high-speed counting, positioning, etc.).

Module	Function
Rack	Support that holds all modules
Power Supply (PS)	Provides electrical power to the modules
Digital I/O	On/Off inputs and outputs (0/1 : button, motor ON/OFF)
Analog I/O	Continuous inputs/outputs (temperature, speed, pressure)
Communication modules	Connection with other PLCs or an industrial network
Function modules	Special functions (high-speed counting, positioning, etc)

Programming Software

- **Siemens:** Step5 (older PLCs), Step7 (S7-300/400), Microwin (S7-200), TIA Portal (new integrated platform for S7-1200/1500 and others).
- **Schneider:** PL7 (older models), Unity Pro (for M340/M580).

16 Criteria for Choosing a PLC

The criteria for selecting a PLC can vary depending on the individual and the project. The choice of a PLC will mainly depend on the specifications and technical requirements of the project. Most PLC manufacturers offer a range of PLCs to suit a variety of projects. Below are some criteria to consider when choosing a PLC:

a. Cycle Time:

Some PLCs have faster cycle times compared to others, making them better suited for systems requiring quick responsiveness.

b. Cost:

For small automation projects, it is preferable to choose micro or mini-PLCs.

c. Development Software:

Some PLCs come with development software equipped with tools for program simulation, saving time during the development phase.

d. Availability of Native Communication Interfaces:

The PLC should be able to communicate with other systems. It is preferable to choose a PLC that supports this communication natively, avoiding the need for additional modules.

e. Availability of Replacement Components:

Some PLC brands have greater market presence, allowing them to offer better after-sales services and customer support.